

BUSINESS REPORT - INFERENTIAL STATISTICS **PROJECT**

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TABLE OF CONTENTS

SNO	CONTENTS
1.	INTRODUCTION TO INFERENTIAL STATISTICS
2.	MODULES DEVELOPED
3.	DEFINING SOME KEY CONCEPTS IN PROBABILITY
4.	MARGINAL PROBABILITY
5.	PROBLEM 1
6.	1.1 Rubric question
7.	1.2 Rubric question
8.	1.3 Rubric question
9.	1.4 Rubric question
10.	Normal distribution
11.	PROBLEM 2
12.	2.1 Rubric question
13.	2.2 Rubric question
14.	2.3 Rubric question
15.	2.4 Rubric question
16.	HYPOTHESIS TEST
17.	PROBLEM 3
18.	3.1 Rubric question
19.	3.2 Rubric question
20.	ANOVA TEST
21.	PROBLEM 4
22.	4.1 Rubric question
23.	4.2 Rubric question
24.	4.3 Rubric question
25.	4.4 Rubric question

FIGURES USED IN THE REPORT:

Figure 1	Data of Football Players types and being injured
Figure 2	Standard Normal Bell Curve
Figure 3	Gunny bag strength of data points $< 3.17\text{kg/sq.cm}$
Figure 4	Gunny bag strength of data points $> 3.6\text{kg/sq.cm}$
Figure 5	Gunny bag strength between 5 and 5.5kg/sq.cm
Figure 6	Bag strength not between 3 to 7.5kg/sq.cm
Figure 7	First five rows of the Zingaro data
Figure 6:	Five- point summary of Zingaro data
Figure 9:	Brief view of Dental Hardness data
Figure 10:	Interaction plot of Method, Dentist, Response

INTRODUCTION TO INFERENTIAL STATISTICS:

Definition: It is a branch of statistics that deals with population or datasets and allows a statistician or a data researcher to make predictions or inferences about the data based on a sample data, and further interpret the results over the large data.

The foundations of inferential statistics start with fundamentals of probability distributions.

Probability(P): The chance or likelihood of an event(outcome) to occur when an experiment or research is conducted.

Formula of Probability:

$$\text{Probability(P) of 'A' event} = m/n$$

m = no of ways the event can happen

n = no of outcomes

Probabilities of Data can be understood in various scenarios by using below modules of Python and data can be further interpreted to get the inferences or insights and formulate the business decisions out of these insights. These insights show strong statistical evidence which becomes pivotal for business managers or key stakeholders to arrive at a valid conclusion.

MODULES DEVELOPED:

Python libraries used	Environment
Numpy, Pandas, Matplotlib, Seaborn, Scipy.stats	Jupyter notebook

DEFINING SOME KEY CONCEPTS IN PROBABILITY:

Marginal Probability:

It is the contingency table (2x2 table with rows and columns) with various numbers or data. Now understanding and analysing the probability of each of these values in comparison to the total value is called as Marginal probability analysis.

We are given a set of Rubric questions relevant to marginal probability which are solved as per below methods in problem 1.

PROBLEM 1:

Problem statement:

A physiotherapist with a male football team is interested in studying the relationship between foot injuries and the positions at which the players play from the data collected.

	Striker	Forward	Attacking Midfielder	Winger	Total
Players Injured	45	56	24	20	145
Players Not Injured	32	38	11	9	90
Total	77	94	35	29	235

Figure 1: Data of Football Players types and being injured

Based on the above data, answer to the following questions is below:

1.1 What is the probability that a randomly chosen player would suffer an injury?

- The data represents a Contingency table which consists of rows and columns of attributes
- Formula to calculate the probability is to calculate the proportion of total number of injured players to the total number of players.
- Players injured = 145
- Total players = 235

Probability is calculated as:

$$P(\text{injury}) = \frac{\text{Players injured}}{\text{Players total}} = \frac{145}{235} = 0.617 \text{ Or } 61.7\%$$

1.2 What is the probability that a player is a forward or a winger?

- To find the probability that a player is a forward or a winger, we sum up both the number of forward and winger players and divide by total players.
- Forward players = 94
- Winger players = 29
- Total players = 235

Probability is calculated as:

$$P(\text{Forward or Winger}) = \frac{\text{forward} + \text{winger}}{\text{total players}} = \frac{94 + 29}{235} = 0.523 \text{ or } 52.3\%$$

1.3 What is the probability that a randomly chosen player plays in a striker position and has a foot injury?

- To find the probability of an injured striker, we divide 45 with total players.
- Injured strikers = 45
- Total players = 235

Probability is calculated as:

$$P(\text{Injured striker}) = \frac{\text{Injured strikers}}{\text{Total players}} = \frac{45}{235} = 0.1914 \text{ or } 19.14\%$$

1.4 What is the probability that a randomly chosen injured player is a striker?

- To find the probability of an injured striker, we divide 45 with total injured players.
- Injured strikers = 45
- Total injured players = 145

Probability is calculated as:

$$P(\text{Striker}|\text{Injured}) = \frac{\text{Injured strikers}}{\text{Total injured players}} = \frac{45}{145} = 0.310 \text{ or } 31\%$$

Normal distributions:

Normal distribution is the most common and the useful distribution.

It is the representation of data in the shape of a bell curve which has two main parameters, mean(μ) and standard deviation(σ)

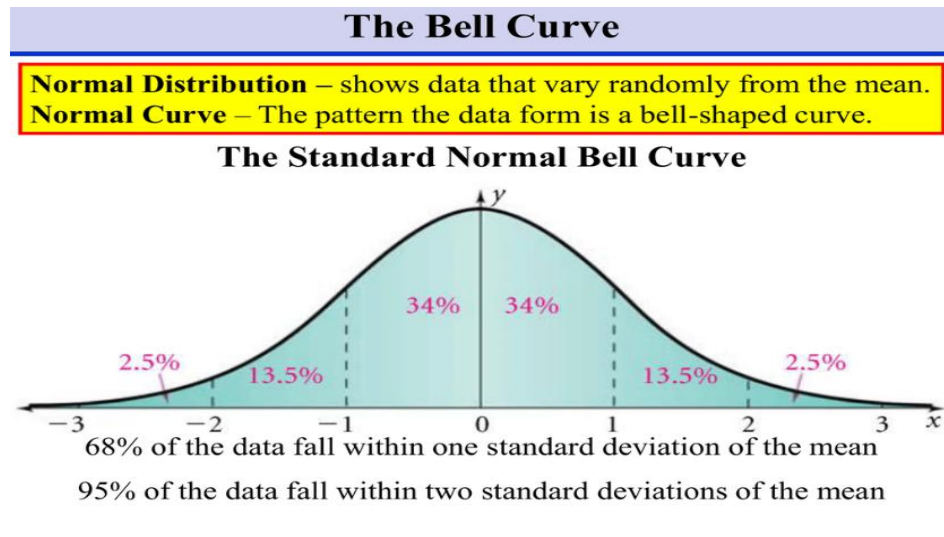


Figure 2: Standard Normal Bell Curve

Source: <https://www.vroque.co/post/bell-shaped-curve-normal-distribution-in-statistics>

It follows an Empirical rule, 68-95-99,

where 68% of data will fall within one standard deviation from the mean,

95% data falls within two standard deviations from the mean and

99% of data falls within three standard deviations from the mean.

Properties:

- Normal distribution is symmetric around the mean
- Mean, median and mode of the normal distribution are equal
- And total area of this bell curve is 1.

The Normal distribution is understood based on below set of question from Problem 2.

PROBLEM 2

Problem statement:

The breaking strength of gunny bags used for packaging cement is normally distributed with a mean of 5 kg per sq. centimetre and a standard deviation of 1.5 kg per sq. centimetre. The quality team of the cement company wants to know the following about the packaging material to better understand wastage or pilferage within the supply chain; Answer the questions below based on the given information; (Provide an appropriate visual representation of your answers, without which marks will be deducted)

$\mu=5$ and $\sigma=1.5$

Answer to the following questions is below:

Z SCORE:

This is another key concept in inferential statistics and probability functions which helps understand how far the data points lie from the mean with respect to the standard deviation.

2.1 What proportion of the gunny bags have a breaking strength of less than 3.17 kg per sq cm?

- To calculate the proportion of gunny bags with breaking strength less than 3.17kg(x), we need to calculate the Z-score of the data.
- Z-score defines how far the data point (i.e., 3.17kg) is from the mean w.r.t standard deviation.
- $x = 3.17$

$$Z\ Score = \frac{x - \mu}{\sigma} = \frac{3.17 - 5}{1.5} = -1.22$$

Insights:

Z score of -1.22 is 11.12%.

- This means 11.12% of the gunny bags have a breaking strength of less than 3.17kg/sq.cm in the data.
- This data can be visually represented as below.
- We have used norm and fill between functions to visually represent the data
- The red area in the graph indicates the proportion of data below 3.17kg/sq.cm.

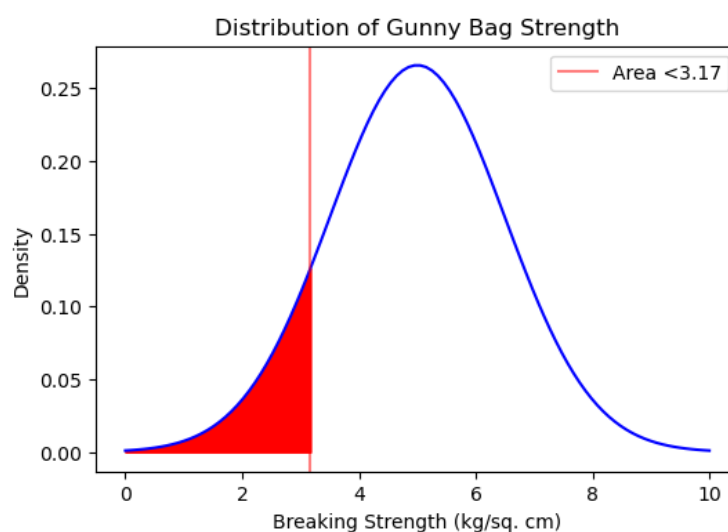


Figure 3: Gunny bag strength of data points <3.17kg/sq.cm

2.2 What proportion of the gunny bags have a breaking strength of at least 3.6 kg per sq cm.?

- To calculate the proportion of gunny bags with breaking strength of at least 3.6kg/sq.cm(x), we need to calculate the Z-score of the data.
- Z-score defines how far the data point (i.e., 3.6kg) is from the mean w.r.t standard deviation.
- $x = 3.6$

$$Z \text{ Score} = \frac{x - \mu}{\sigma} = \frac{3.6 - 5}{1.5} = -0.93$$

Insights:

Z score of -0.93 is 17.61%.

- Since we are calculating probability of breaking strength at least 3.6kg/sq.cm, we are looking at data on and above 3.6kg/sq.cm.
- So, we calculate $1.00 - 0.17 = 0.82$
- This means 82% of the total gunny bags have a breaking strength of at least 3.6kg/sq.cm in the data.
- This data can be visually represented as below.

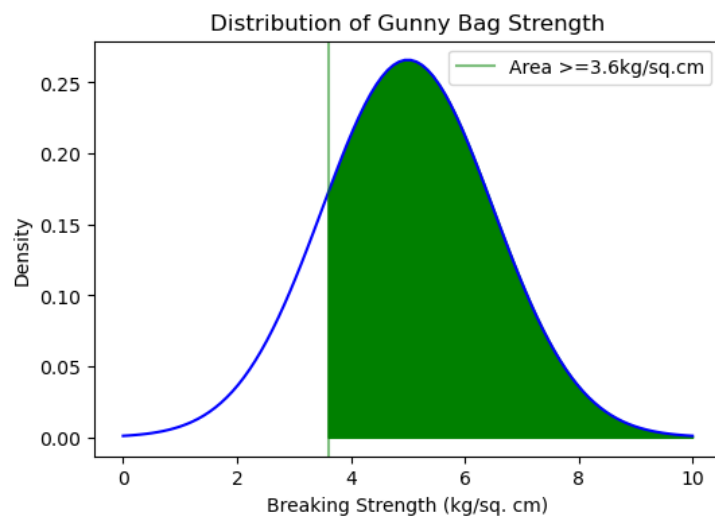


Figure 4: Gunny bag strength of data points $>3.6\text{kg/sq.cm}$

2.3 What proportion of the gunny bags have a breaking strength between 5 and 5.5 kg per sq cm.?

- To calculate the proportion of gunny bags with breaking strength between 5 and 5.5kg/sq.cm, we need to calculate the Z-score of both the datapoints.
- $(\mu, \sigma) = (5, 1.5)$
- $X_1 - 5$
- $X_2 - 5.5$
- We want to know the $P(5 \leq X \leq 5.5)$

$$Z1(\leq 5) = \frac{x1 - \mu}{\sigma} = \frac{5 - 5}{1.5} = 0. \text{Hence, } Z1 = 0.5$$

$$Z2(\geq 5.5) = \frac{x2 - \mu}{\sigma} = \frac{5.5 - 5}{1.5} = 0.3333. \text{Hence, } Z2 = 0.63$$

- Final Z score is $0.63 - 0.5 = 0.1306$ or 13.06%

Insights:

- Z score of 13.06% which means 13.06% of the gunny bags have a breaking strength between 5 and 5.5 kg per sq.cm.
- This data can be visually represented as below.

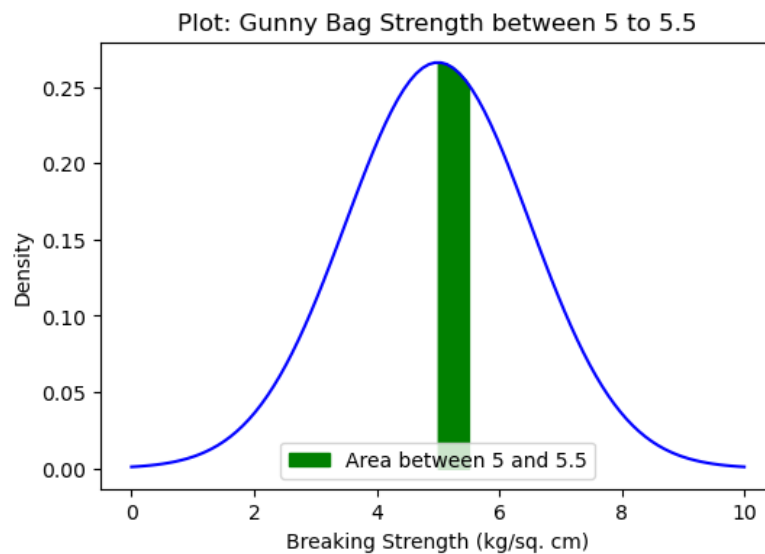


Figure 5: Gunny bag strength between 5 and 5.5kg/sq.cm

2.4 What proportion of the gunny bags have a breaking strength NOT between 3 and 7.5 kg per sq.cm.?

- To calculate the proportion of gunny bags with breaking strength NOT between 3 and 7.5kg/sq.cm, we need to calculate the Z-score of both the datapoints, subtract the z scores, and further deduct the value from 1.
- $(\mu, \sigma) = (5, 1.5)$
- $X1 = 3$
- $X2 = 7.5$
- We want to know the $P(3 \leq X \leq 7.5)$

$$Z1(\leq 3) = \frac{x1 - \mu}{\sigma} = \frac{3 - 5}{1.5} = -1.3333. \text{Hence, } Z1 = -0.0918$$

$$Z2(\geq 7.5) = \frac{x2 - \mu}{\sigma} = \frac{7.5 - 5}{1.5} = 1.6667. \text{Hence, } Z2 = 0.9525$$

Final Z score is : $0.9525 - 0.0918 = 0.8607$ or 86.07%

- The proportion of gunny bags NOT between 3 and 7.5 g/sq.cm:

$$\begin{aligned} 1 - P(3 \leq X \leq 7.5) &= 1 - 0.8607 = 0.13931 - P(3 \leq X \leq 7.5) \\ &= 1 - 0.8607 = 0.1393 \text{ or } 13.93\% \end{aligned}$$

Insights:

- Z score of 13.93% which means 13.93% of the gunny bags have a breaking strength NOT between 3 and 7.5 kg per sq.cm.
- This data can be visually represented as below.

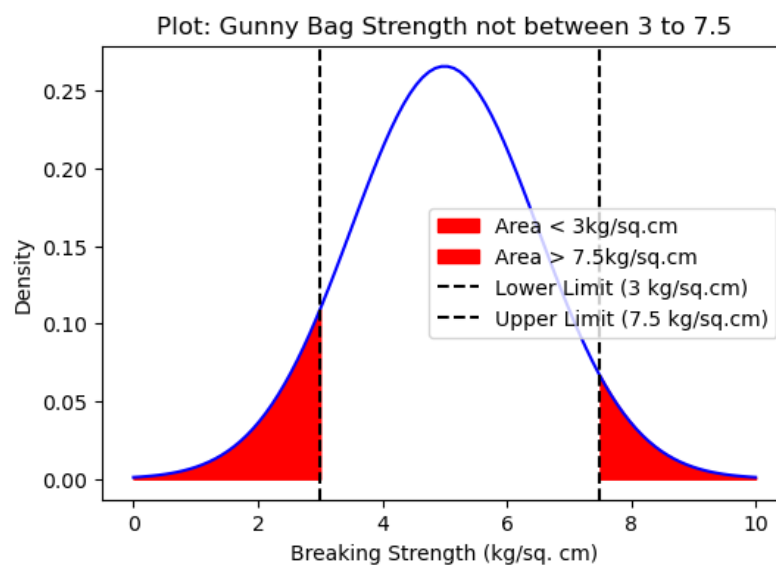


Figure 6: Bag strength not between 3 to 7.5kg/sq.cm

HYPOTHESIS TEST:

This is a method of statistics used to decide whether the data provide sufficient evidence to reject a particular hypothesis. This hypothesis test usually involves a null hypothesis and an alternative hypothesis. The decision is usually made based on comparing the test statistic to a p-value.

P-value means a probability value that is a value which describes how likely we can find the set of datapoints if the null hypothesis is true. The p-value if is >0.05 significance level, we usually reject the null hypothesis and if p-value is >0.05 significance level, we fail to reject null hypothesis.

Significance level is a way of saying that the p-value of a statistical test is small enough to reject the null hypothesis of the test.

Let us look at the below problem relevant to various hypothesis tests and solve them with the suitable test using python.

PROBLEM 3:

Problem statement:

Zingaro stone printing is a company that specializes in printing images or patterns on polished or unpolished stones. However, for the optimum level of printing of the image, the stone surface must have a Brinell's hardness index of at least 150. Recently, Zingaro has received a batch of polished and unpolished stones from its clients. Use the data provided to answer the following (assuming a 5% significance level);

Problems to be solved:

3.1 Zingaro has reason to believe that the unpolished stones may not be suitable for printing. Do you think Zingaro is justified in thinking so?

3.2 Is the mean hardness of the polished and unpolished stones the same?

Note: Consider a 5% significance level for the analysis

Data observations:

- We import the data set into python and observe the data with the help of head, shape, info, describe functions.
- We can understand the data types, five-point statistical summary of the data, no of rows and columns in the data and understand a brief of the data with these functions.
- Based on the observations, we check for missing values and remove them using is-null function.

	Unpolished	Treated and Polished
0	164.481713	133.209393
1	154.307045	138.482771
2	129.861048	159.665201
3	159.096184	145.663528
4	135.256748	136.789227

Figure 7: First five rows of the Zingaro data

	count	mean	std	min	25%	50%	75%	max
Unpolished	75.0	134.110527	33.041804	48.406838	115.329753	135.597121	158.215098	200.161313
Treated and Polished	75.0	147.788117	15.587355	107.524167	138.268300	145.721322	157.373318	192.272856

Figure 8: Five- point summary of Zingaro data

3.1. Zingaro has reason to believe that the unpolished stones may not be suitable for printing. Do you think Zingaro is justified in thinking so?

Solution:

The Null and alternative hypotheses can be formulated as:

Null hypothesis: Unpolished stones are suitable for printing.

Alternative hypothesis: Unpolished stones are not suitable for printing.

Assumptions:

- Data is assumed to be normally distributed
- Samples are independent
- Samples show unequal variances

Hypothesis test can be used: Levene's test for variance:

- Since the question is about checking the suitability of both the type of stones for printing, and when we see the variances or standard deviations are different, we can check how significantly they are different using the Levene's test.
- It can be conducted by importing levene function from stats module of python.

Insights:

- P-Value is: 3.4771056592659467e-07
- since p-value is much less than 0.05 significance we can statistically infer that unpolished stones may not be suitable for printing (Significance<0.05)

3.2 Is the mean hardness of the polished and unpolished stones the same?

Solution:

The Null and alternative hypotheses can be formulated as:

Null hypothesis: $\mu_1 = \mu_2$ - Mean hardness of polished stones is same as unpolished stones.

Alternative hypothesis: $\mu_1 \neq \mu_2$ - Mean hardness of polished stones is different to unpolished stones.

Assumptions:

- Samples are independent - unpolished and polished
- Normal distributed
- Homogenous variances
- Continuous data

Hypothesis test can be used: Two sample T Test:

- Two sample T Test is used here to compare two independent groups and their means.
- It can be conducted through `ttest_ind` function from stats module of python.

Insight:

- P-value: 0.001588379295584306
- Inference: Mean hardness of polished stones is different to unpolished stones.
- As the p-value is 0.0015 which is much lower than the level of significance, so We reject null hypothesis. Hence, we have enough statistical evidence to say that hardness of unpolished stones is not same as polished stones.

ANOVA TEST – ANALYSIS OF VARIANCE:

Analysis of Variance (ANOVA) is used to test the difference between the means of two or more groups.

ANOVA helps analyse the heterogeneity within the groups as well between the groups.

An F-test returns the ANOVA value.

Types of ANOVA:

One – Way ANOVA:

This test is used to see if there is a variation in the mean values of three or more groups.

It is used if the data has only one independent variable.

Two – Way ANOVA:

Two independent variables are used in the two-way ANOVA. It compares two or more independent variables and analyses the influence within the variables and among the variables. This interaction is termed as interaction effect.

Let us look at a problem and analyse the solution using a ANOVA test.

PROBLEM 4:

Problem statement:

Dental implant data: The hardness of metal implants in dental cavities depends on multiple factors, such as the method of implant, the temperature at which the metal is treated, the alloy used as well as the dentists who may Favor one method above another and may work better in his/her favourite method. The response is the variable of interest.

Problems to be solved:

- 4.1 How does the hardness of implants vary depending on dentists?
- 4.2 How does the hardness of implants vary depending on methods?
- 4.3 What is the interaction effect between the dentist and method on the hardness of dental implants for each type of alloy?
- 4.4 How does the hardness of implants vary depending on dentists and methods together?

Note: Consider a 5% significance level for the analysis.

Data observations:

- We import the data set into python and observe the data with the help of head, shape, info, describe functions.

- We can understand the data types, five-point statistical summary of the data, no of rows and columns in the data and understand a brief of the data with these functions.
- Based on the observations, we check for missing values and remove them using is-null function.

	Dentist	Method	Alloy	Temp	Response
0	1	1	1	1500	813
1	1	1	1	1600	792
2	1	1	1	1700	792
3	1	1	2	1500	907
4	1	1	2	1600	792
..	...	...	...	...	...
85	5	3	1	1600	483
86	5	3	1	1700	405
87	5	3	2	1500	536
88	5	3	2	1600	405
89	5	3	2	1700	312

[90 rows x 5 columns]

Figure 9: Brief view of Dental Hardness data

Data Dictionary:

- Dentist - 5 dentists are there to do the treatments.
- Method - 3 methods are used.
- Alloy - 2 Different types of alloys are used.
- Temp - Temperature at which the metal is treated.
- Response - It is the variable of interest.

4.1 How does the hardness of implants vary depending on dentists?

Let's write the null and alternative hypothesis

Null hypothesis: Mean hardness of implants does not depend on dentists

Alternative hypothesis: Mean hardness of implants depends on dentists

Checking the assumptions:

- Assumption1 - For testing of normality, Shapiro-Wilk's test is applied to the response variable.
- Assumption2 - For equality of variance, Levene test is applied to the response variable.
- Assumption3 – Data to be normally distributed data

Assumption1 - Shapiro-Wilk's test:

Let's write the null and alternative hypothesis:

Null hypothesis: The data follows a normal distribution

Alternative hypothesis: The data does not follow a normal distribution

The shapiro function is used to calculate the p-value.

Insights:

- The p-value is 8.080212865024805e-06
- Inference: The p-value is much less than 0.05% significance. hence we reject the null hypothesis and say sample is not normally distributed.

Assumption2 - Levene's test:

Let's write the null and alternative hypothesis:

Null hypothesis: The variance of hardness amongst various dentists is equal

Alternative hypothesis: The variance of hardness amongst various dentists is not equal

The levene function is used to calculate the p-value.

Insights:

- p-value is: 0.007858817382355401
- Variances are significantly different as the p-value is less than 0.05% significance.

Validating the assumptions to perform one way ANOVA:

- Assumption1 - Shapiro-Wilk's test says the data is not normally distributed.
- Assumption2 - Levene test says variances are significantly different.
- Assumption3 - Sample is independent simple random sample

Note: As per the rubric I am proceeding with performing the further test even if the assumptions of the test fail.

Insights:

- p-value is: 0.007858817382355401
- We reject null hypothesis and at least one dentist has a significantly different mean hardness
- Since the mean hardness is affected by at least one dentist, we proceed and perform multiple comparison test to understand which dentist is showing the variance.
- This can be studied by performing a Tukey HSD test as below.

Multiple Comparison test (Tukey HSD):

Defining Null and Alternate hypothesis:

Null hypothesis: Mean Implant hardness of one dentist does not differs to others

Alternate hypothesis: Mean Implant hardness of one dentist differs to others

From stats models we use the function pairwise tukey hsd test to perform this analysis.

Insights:

- Since reject column reads False, we say that there is no significant difference in mean implant hardness between the dentists.
- A p-value of less than 0.05 indicates statistically significant differences. However, we see no values less than 0.05, which once again supports the above statement as no difference in implant hardness between the dentists.

4.2 How does the hardness of implants vary depending on methods?

Let's write the null and alternative hypothesis

Null hypothesis: Variance of implants hardness is equal across different methods

Alternate hypothesis: Variance of implants hardness differs across different methods

Checking the assumptions:

- Assumption1 - For testing of normality, Shapiro-Wilk's test is applied to the response variable - this is already performed in 4.1 question
- Assumption2 - For equality of variance, Levene test is applied to the response variable 'Methods'.
- Assumption3 - Normally distributed data

Performing levene test:

- P-value for levene test is: 0.004138452940152019
- Variances are significantly different across methods (<0.05).
- Since the p-value is less than the 0.05 significance level, we reject the null hypothesis and variances are significantly different across methods.

Performing One-way ANOVA test:

- Let us proceed and conduct one way ANOVA test on the data and compute p-value using F-oneway function.
- P-value for ANOVA test is: 7.683891892977992e-08
- Since p value is extremely small, we reject the null hypothesis and at least one method has a significantly different mean hardness.

Performing a Two-way ANOVA:

- However, we do not know which method is different from the other methods or whether all methods are different. Multiple comparison tests are used to test the differences between all pairs of means of these methods.
- Pairwise-tukey hsd test is used to do the multiple comparison tests on this data i.e., a two-way Anova test is conducted.

- The results infer that for group1 response and group2 from dentist show a p-value greater than 0.05 hence we fail to reject null hypothesis and say implant hardness variance is similar across methods for this pair.
- Both the other pairs show a 0 p-value which indicates hardness varies significantly across methods for these pairs.

4.3 What is the interaction effect between the dentist and method on the hardness of dental implants for each type of alloy?

- Since we are asked to compare the mean hardness of implants between two factors, i.e., dentist and method with respect to each type of alloy, it is a two-way ANOVA test.
- The pairwise- tukey hsd function of stats will be used to compute the test statistic and p-value.

Let us write the null and alternative hypothesis:

Null hypothesis: No significant interaction effect between dentists and methods on the implants hardness for alloy type

Alternate hypothesis: Significant interaction effect observed between dentists and methods on the implants hardness for alloy type

Ols - Ordinary least squares method is used to reduce the sum of squares of the predicted data.

Insights:

- P value less than 0.05 for Dentist: method indicates a significant interaction seen for Alloy type 1.
- P value more than 0.05 for Dentist: method indicates no significant interaction seen for Alloy type 2.
- Said data is plotted as per below graph.

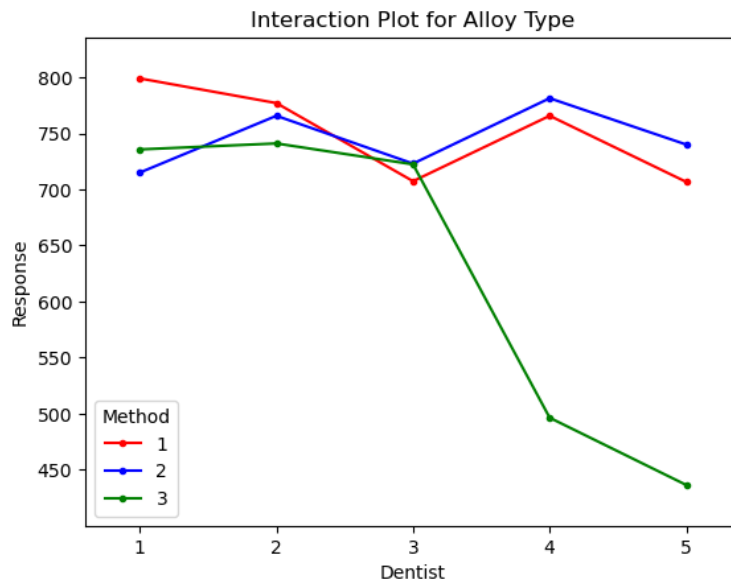


Figure 10: Interaction plot of Method, Dentist, Response

Insights:

- Method 3 shows strong interaction with the dentist, as graph line drops for Dentists 4 and 5.
- This may indicate that Method 3 is highly sensitive to individual dentist techniques, while Methods 1 and 2 is more stable with dentists.

4.4 How does the hardness of implants vary depending on dentists and methods together?

- Since we are asked to compare the mean hardness of implants between two factors, i.e., dentist and method together, it is a two-way ANOVA test.
- The pairwise tukey hsd function of stats models will be used to compute the test statistic and p-value

Let us write the null and alternative hypothesis:

Null hypothesis: No significant interaction effect between dentists and methods with respect to alloy type

Alternate hypothesis: Significant interaction effect observed between dentists and methods with respect to alloy type

Checking the assumptions:

- Assumption1 - For testing of normality, Shapiro-Wilk's test is applied to the response variable - this is already performed in 4.1 question
- Assumption2 - For equality of variance, Levene test is applied to the response variable 'Methods' and 'Dentist' which is already performed in the 4.2 question
- Assumption3 - Normally distributed data

Insight:

- 2-8, 3-8,4-8,5-8 do not show effects as the p value is high, so dentist and method may not impact hardness for these combinations.
- 2-3,2-7,4-7 show low p-values, which indicates dentist and method influence the hardness.
- This suggests that dentist and some methods together influence hardness largely when compared to some other methods.